

Tribhuvan University
Institute of Science and Technology
2066

Bachelor Level/ First Year/ First Semester/ Science
Computer Science and Information Technology (MTH:104)
(Calculus and Analytical Geometry)

Full Marks: 80
Pass Marks: 32
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all questions.

Group A (10×2=20)

1. Find the length of the curve $y = x^{3/2}$ from $x=0$ to $x=4$.
2. Find the critical points of the function $f(x) = x^{3/2}(x-4)$.
3. Does the following series converge?
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$
4. Find the polar equation of the circle $(x+2)^2 + y^2 = 4$
5. Find the area of the parallelogram where vertices are A(0,0), B(7,3), C(9,8) and D(2,5).
6. Evaluate the integral $\int_0^{2t} \int_0^1 (\sin x + \cos y) dx dy$
7. Evaluate the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$
8. Find $\left(\frac{\partial w}{\partial x}\right)_{y,z}$ if $w = x^2 + y - z + \sin t$ and $x + y = t$.
9. Solve the partial differential equation $p + q = x$.
10. Find the general integral of the linear partial differential equation $z(xp - yq) = z^2 - x^2$.

Group B (5×4=20)

11. State and prove Rolle's Theorem.
12. Find the length of the cardioid $r = 1 + \cos\theta$.
13. Define unit tangent vector of a differentiable curve. Find the unit tangent vector of the curve $r(t) = (\cos t + t \sin t)i + (\sin t - t \cos t)j$, $t > 0$.
14. What do you mean by critical point of a function $f(x,y)$ in a region? Find local extreme values of the function $f(x,y) = xy - x^2 - y^2 - 2x - 2y + 4$.
15. Find a particular integral of the equation

$$\frac{\partial^2 z}{\partial x^2} - \frac{\partial z}{\partial y} = 2y - x^2$$

Group C (5×8=40)

16. Graph the function $y = x^{4/3} - 4x^{1/3}$

17. What do you mean by Taylor's polynomial of order n ? Obtain Taylor's polynomial and Taylor's series generated by the function $f(x) = \cos x$ at $x = 0$.

18. Find the volume of the region enclosed by the surface $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$.

19. Obtain the absolute maximum and minimum values of the function.

$f(x, y) = 2 + 2x + 2y - x^2 - y^2$ on the triangular plate in the first quadrant bounded by lines $x = 0, y = 0, y = 9 - x$.

OR

Evaluate the integral $\int_0^1 \int_0^{3-3x} \int_0^{3-3x-y} dz dy dx$

20. Show that the solution of the wave equation $\frac{d^2 u}{dt^2} = c^2 \frac{d^2 u}{dx^2}$, $c^2 = \frac{T}{\rho}$, is $u(x, t) =$

$\frac{1}{2} [f(x + ct) + f(x - ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$ and deduce the result if the velocity is zero.

OR

Find a particular integral of the equation $(D^2 - D^1) = A \cos(lx + my)$ where A, l, m are constants.